

## PART-1 : MATHEMATICS

### SECTION-I (i)

1) Which of the following are orthogonal curves

- (A)  $xy = 2$  and  $x^2 - y^2 = 3$   
 (B)  $y = e^x$  and  $y = e^{-x}$   
 (C)  $y = x^2$  and  $x^2 + 2y^2 = 3$   
 (D)  $\frac{x^2}{4} - \frac{y^2}{3} = 1$  and  $\frac{x^2}{16} + \frac{y^2}{9} = 1$

2) Let  $y = f(x)$  be a non-zero real valued differentiable function  $\forall x \in \mathbb{R}$ , such that  $\frac{f'(x)}{f(x)} \neq -1$  and  $(f'(x))^3 + x^3 + 3xf(x)f'(x) = (f(x))^3 \forall x \in \mathbb{R}$ , then which of the following statements is(are) TRUE ?

- (A) if  $f(-1) = 0$ , then  $f(x) = (x + 1)e^x$   
 (B) if  $f(-1) = 1$ , then  $f(x) = (x + 1)e^{(x+1)}$   
 (C)  $\frac{f'(c) - f(c)}{c} = -1$  for some  $c \in \mathbb{R}$   
 (D)  $\frac{f'(c) - f(c)}{c} = e$  for some  $c \in \mathbb{R}$

3) Let  $f(x)$  be a differentiable function on  $[0, 8]$  such that  $f(1) = 5, f(2) = \frac{1}{2}, f(3) = 6, f(4) = -3, f(5) = 4, f(6) = \frac{1}{4}$  and  $f(7) = \frac{-1}{5}$ . If the minimum number of roots of the equation  $f'(x) - f'(x)(f(x))^2 = 0$  is  $\lambda$  and that of  $2.f(x).f'(x) + (f'(x))^2 + f(x).f''(x) = 0$  is  $\mu$ , then

- (A)  $\lambda + \mu = 16$   
 (B)  $\lambda - \mu = 5$   
 (C)  $\lambda - \mu = 6$   
 (D)  $\lambda + \mu = 17$

$$\alpha = \sum_{k=1}^{\infty} \sin^{2k} \left( \frac{\pi}{6} \right).$$

4) Let

Let  $g : [0, 1] \rightarrow \mathbb{R}$  be the function defined by

$$g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$$

Then, which of the following statements is/are TRUE ?

- (A) The minimum value of  $g(x)$  is  $2^{\frac{7}{6}}$

- (B) The maximum value of  $g(x)$  is  $1 + \frac{1}{2^{\frac{1}{3}}}$
- (C) The function  $g(x)$  attains its maximum at more than one point
- (D) The function  $g(x)$  attains its minimum at more than one point

5) Let  $f(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + e$  (where  $a, b, c, d, e \in \mathbb{R}$ ) be a polynomial. A line 'L' touches the curve  $y = f(x)$  at two point P(1, 2) and Q(4, 2). If Q is the point of inflection for  $y = f(x)$  then choose the correct option(s) :-

- (A) the value of  $f(2) + f(3)$  is '-8'
- (B) the value of  $f(2) + f(3)$  is '8'
- (C) at  $x = \frac{11}{5}$ ,  $f(x)$  has a local minimum
- (D) at  $x = \frac{11}{5}$ ,  $f(x)$  has a local maximum

6) Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be given by

$$f(x) = \begin{cases} x^7 + 7x^6 + 21x^5 + 35x^4 + 35x^3 + 19x^2 + 7x + 1 & , \quad x < 0; \\ x^2 - x + 1 & , \quad 0 \leq x < 1; \\ \frac{2}{3}x^3 - 4x^2 + 7x - \frac{8}{3} & , \quad 1 \leq x < 3 \\ (x-2) \log_e (x-2) - x + \frac{10}{3} & , \quad x \geq 3 \end{cases}$$

Which of the following options is/are correct ?

- (A)  $f(x)$  has a local maximum at  $x = 1$
- (B)  $f(x)$  is onto
- (C)  $f(x)$  is increasing on  $(-\infty, 0)$
- (D)  $f(x)$  is NOT differentiable at  $x = 1$

#### SECTION-I (ii)

#### Common Content for Question No. 1 to 2

Consider a sequence  $\{a_n\}$  such that  $a_1 = 0$  and  $\frac{1}{1-a_{n+1}} - \frac{1}{1-a_n} = 2n+1$  ( $n = 1, 2, 3, \dots$ ). Also

$$b_k = \sqrt{\frac{k+1}{k}} (1 - \sqrt{a_{k+1}}) \quad (k = 1, 2, 3, \dots)$$

On the basis of above information, answer the following questions :

1)  $\prod_{n=2}^{\infty} a_n$  is equal to -

- (A)  $\frac{1}{2}$
- (B) 1

(C)  $\frac{3}{4}$

(D) infinite

2)  $\lim_{n \rightarrow \infty} \sum_{k=1}^n b_k$  is equal to -

(A)  $\sqrt{2}$

(B) 2

(C)  $\sqrt{2} - 1$

(D)  $\sqrt{2} + 1$

### Common Content for Question No. 3 to 4

Consider the number sequence  $a_n = \frac{1}{(n+1)\sqrt{n} + n\sqrt{n+1}}$ , where  $n$  is a positive integer. The sum of the first  $n$  term is given by  $S_n$ .

3) How many terms in  $S_1, S_2, S_3, S_4 \dots S_{2014}$  are rational numbers -

(A) 42

(B) 43

(C) 44

(D) 41

4) Let  $T_n = (1 - S_n)^{-2}$ , the value of  $T_1 + T_2 + T_3 + \dots + T_{10}$  is equal to -

(A) 65

(B) 56

(C) 11

(D) 55

### SECTION-II (i)

### Common Content for Question No. 1 to 2

Let  $f$  be a function defined on  $\mathbb{R}$  (set of all real numbers) such that  $f'(x) = 2018(x - 2019)^3(x - 2020)^4(x - 2021)^5(x - 2022)^6$  for all  $x \in \mathbb{R}$ . If  $g$  is a positive valued function defined on  $\mathbb{R}$  such that  $f(x) = \int n(g(x))$ , for all  $x \in \mathbb{R}$ .

1) Then the number of points in  $\mathbb{R}$  at which  $g(x)$  has a local minima, is

2) Let  $h(x) = f'(x) \left( \frac{g''(x)}{g(x)} - \left( \frac{g'(x)}{g(x)} \right)^2 \right)$ , then minimum number of distinct real roots of  $h'(x) = 0$  is \_\_\_\_\_

### Common Content for Question No. 3 to 4

Let  $f(x)$  be a continuous and differentiable function satisfying the following conditions

$$\prod_{r=1}^6 f(r) < 0, \quad \prod_{r=1}^3 f(2r) < 0, \quad \prod_{r=1}^2 f(3r) < 0$$

(a)  $f(6) > 0$  and

(c)  $f(x)$  is monotonic in  $(n, n + 1)$ ,  $n \in \mathbb{I}$

3) Let A denotes the set consisting of number of distinct possible roots of  $f(x) = 0$  in  $x \in (1, 6)$ . Find the sum of all the elements of set A.

4) Let B denotes the set consisting of min. number of distinct possible real roots of  $f'(x) = 0$  in  $x \in (1, 6)$ . Find number of elements in set B.

### Common Content for Question No. 5 to 6

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function defined by  $f(x) = -x^3 - 3x^2 - 6x + 1$ .

5) Number of integers in the solution set of  $x$  satisfying the inequality  $f(f(x^3 + f(x))) \geq f(f(-f(x) - x^3))$  is

6) Number of real roots of equation  $f(f(x^2)) = 0$  is

### SECTION-II (ii)

$$N = \frac{\overbrace{999\dots9}^{16 \text{ digits}} - \overbrace{5555\dots5}^{8 \text{ digits}}}{\underbrace{9000\dots04}_{9 \text{ digits}}}$$

1) Let , then sum of digits of N is

2) The set of all values of ' $\alpha$ ' for which the function  $f(x) = (\alpha^2 - 3\alpha + 2)(\cos 2x) + (\alpha - 1)x + \cos 7$  doesn't have any stationary point, is A, then number of integer is Set A is

3) Let  $x, y \in \mathbb{R}$  such that  $x^2 + y^2 = 4 \forall x, y \in \mathbb{R}$ . Let ' $\alpha$ ' denotes the least possible value of  $\frac{y+2}{x+y-4}$ , then value of  $|\alpha|$  is

## PART-2 : PHYSICS

### SECTION-I (i)

1) A spherical soap bubble of mass  $m = 12\pi \text{ mg} = 12\pi \times 10^{-6} \text{ kg}$  and surface tension  $\sigma = 1.0 \times 10^{-2} \text{ N m}^{-1}$  is inflated through a short thin tube open to the atmosphere, so the pressure difference due to the enclosed gas may be neglected. The bubble carries charge Q and has equilibrium radius  $R_0 = 10 \text{ cm}$ . Assume the charge is uniformly distributed over the soap film. The effective electric field acting

on the surface charge is  $E_0 = \frac{Q}{8\pi\epsilon_0 R^2}$ ,  $\epsilon^0 = 9.0 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$ .

Neglect gravity, air resistance, viscous damping and the inertia of the enclosed air. For small spherical radial oscillations, assume  $Q$  remains constant. The charge is then suddenly increased to  $Q_1 = 10Q$ . Immediately after this increase, the radius is still  $R_0$ . The bubble subsequently breaks into a large number of fine aerosol droplets. Neglect the final surface and electrostatic energies of the droplets and assume that the initial surface and electrostatic energies of the bubble are converted completely into their kinetic energy. Which of the following statements are correct ?

- (A) The initial equilibrium charge is  $Q \simeq 3.4 \times 10^{-7} \text{ C}$ .
- (B) The angular frequency of small radial oscillations is  $\omega = 200 \text{ rad s}^{-1}$ .
- (C) Immediately after the charge becomes  $10Q$ , the outward radial acceleration is  $1.32 \times 10^5 \text{ m s}^{-2}$ .
- (D) After breakup, the estimated speed of the aerosol droplets is  $v \simeq 164 \text{ ms}^{-1}$ .

2) A tall, isolated, electrically neutral metallic cylinder of height  $h$  is placed vertically in a uniform gravitational field  $\vec{g} = -g\hat{z}$ . The  $z$ -axis is taken vertically upward, with the bottom face at  $z = 0$  and the top face at  $z = h$ . The positive ion lattice is fixed, while the conduction electrons are treated as an ideal classical gas at uniform temperature  $T$ . Let  $n(z)$  denote the electron number density and  $p(z) = n(z)k_B T$  the electron pressure. In equilibrium, the electron fluid has zero drift velocity. Neglect edge and fringing effects, so that all quantities depend only on  $z$ . Assume that variations in  $n$  are sufficiently small that  $n \approx n_0$  wherever it appears as a multiplicative factor. The electrons as charge  $-e$  and mass  $m$ . Which of the following statements are correct ?

- (A) The equilibrium condition for the electron gas is  $\frac{dp}{dz} = -n_0 (mg + eE_z)$ .
- (B) If the electron pressure is uniform throughout the conductor, then  $\vec{E} = \frac{m}{e}\vec{g}$ .  
If the internal electric field  $E_z$  is uniform, the electron density varies approximately as
- (C)  $n(z) \approx n(0) - \frac{n_0 (mg + eE_z)}{k_B T} z$ .

- (D) If the electron density remains uniform, the potential difference between the top and bottom faces is  $V_{\text{top}} - V_{\text{bottom}} = \frac{mgh}{e}$ , with the upper face at higher electric potential.

3) A rigid electric dipole of mass  $m$  and dipole moment magnitude  $p$  and negligible size is constrained to move along the  $x$ -axis while remaining free to rotate in the  $xy$ -plane. The external electric field is  $E(x) = E_0 \left(1 + \frac{x}{L}\right)^2 \hat{i}$ , where  $E_0$  and  $L$  are positive constants. At an instant, the dipole is located at  $x = L$ , and its dipole moment makes an angle  $60^\circ$  with the  $+x$ -axis. Neglect higher-order multipole effects. Which of the following statements are correct?

- (A) The magnitude of the torque on the dipole is  $\tau = \sqrt{3}pE_0$ .
- (B) The net force on the dipole is directed along the  $+x$ -axis and has magnitude  $F = \frac{2pE_0}{L}$ .
- (C) The potential energy of the dipole at this instant is  $U = -2pE_0$ .

If the orientation of the dipole is constrained to remain fixed at  $60^\circ$  with the x-axis, its

(D) acceleration at  $x = L$  is  $a = \frac{4pE_0}{mL}$ .

4) A planet moves in an elliptical orbit of semimajor axis  $a$  and eccentricity  $e = \frac{1}{2}$  around a star fixed at one focus. Let P and Q be the two points at which the planet-star distance is  $a$ . The planet moves

from P to Q through aphelion. Let the orbital period be  $T$ , and let  $v_0 = \sqrt{\frac{GM}{a}}$  where  $M$  is the mass of the star. Which of the following statements are correct?

At each of the points P and Q, the radial and transverse velocity components have magnitudes

(A)  $|v_r| = \frac{v_0}{2}$ ,  $v_t = \frac{\sqrt{3}}{2}v_0$ , so the velocity makes an angle  $30^\circ$  with the local transverse direction.

(B) The time taken by the planet to move from P to Q through aphelion is  $\Delta t_{PQ} = \frac{\pi + 1}{2\pi}T$ .

(C) The ratio of the angular speeds of the planet at perihelion and at either P or Q is  $\frac{\dot{\theta}_{\text{peri}}}{\dot{\theta}_P} = 4$ .

The area swept by the radius vector while the planet moves from P to Q through aphelion is

(D)  $\frac{a^2}{2}(\pi + 1)$ .

5) Four infinitely large, parallel conducting plates, A,B,C,D are located respectively at  $x = 0$ ,  $x = d$ ,  $x = 3d$ ,  $x = 6d$ . Plates A and C are connected by an ideal conducting wire, while plates B and D are connected by another ideal conducting wire. The charge and electric-field effects to the connecting wires may be neglected.

The sum of net surface charge per unit area on the connected combination A + C is  $+3\sigma$ , while that on the connected combination B + D is  $-3\sigma$ , where  $\sigma > 0$ . The electric field vanishes outside the entire system. Neglect edge effects.

Let  $\vec{E}_1, \vec{E}_2, \vec{E}_3$  denote the electric fields in the gaps AB, BC, CD, respectively. One or more of the following statements may be correct.

The electric fields in the gaps AB, BC, and CD are, respectively,

(A)  $\vec{E}_1 = \frac{18\sigma}{11\epsilon_0}\hat{i}$ ,  $\vec{E}_2 = -\frac{9\sigma}{11\epsilon_0}\hat{i}$ ,  $\vec{E}_3 = \frac{6\sigma}{11\epsilon_0}\hat{i}$ .

The net surface charge densities on plates A, B, C, and D are, respectively,

(B)  $\sigma_A = \frac{18\sigma}{11}$ ,  $\sigma_B = -\frac{27\sigma}{11}$ ,  $\sigma_C = \frac{15\sigma}{11}$ ,  $\sigma_D = -\frac{6\sigma}{11}$ .

The magnitude of the potential difference between the two connected combinations is

(C)  $\Delta V = \frac{18\sigma d}{11\epsilon_0}$ .

(D) The electrostatic force per unit area on plate C is directed along +x and has magnitude  $\frac{45\sigma^2}{242\epsilon_0}$ .

6) A uniform solid sphere of mass  $M$  and radius  $R$  is surrounded by a concentric thin spherical shell of mass  $2M$  and radius  $2R$ . A point mass  $m$  moves under the gravitational field of this fixed composite body. The point mass can move freely through the material of the sphere and shell, and all non-gravitational interactions are neglected. Take gravitational potential to be zero at infinity.

Which of the following statements are correct ?

- (A) When the point mass is at a distance  $\frac{R}{2}$  from the centre, its gravitational potential energy and the magnitude of the gravitational force acting on it are respectively  $U = -\frac{19GMm}{8R}$ ,  $F = \frac{GMm}{2R^2}$ .  
The quasistatic work done by an external agent in moving the point mass from  $r = \frac{R}{2}$  to  $r = \frac{3R}{2}$  is  $W_{\text{ext}} = \frac{17GMm}{24R}$ .
- (B) The minimum speed required to project the point mass from the centre so that it reaches infinity with zero final speed is  $v_{\text{esc}} = \sqrt{\frac{5GM}{R}}$ .
- (C) If the point mass moves in a circular orbit of radius  $\frac{R}{2}$  inside the solid sphere, its total mechanical energy is  $E = -\frac{17GMm}{8R}$ .
- (D)

#### SECTION-I (ii)

#### Common Content for Question No. 1 to 2

A spacecraft is initially moving in a circular parking orbit at an altitude of 250 km above Earth. Instead of following the minimum-energy Hohmann transfer orbit, it is injected into a heliocentric elliptical trajectory whose perihelion lies at Earth's orbital radius and whose semimajor axis is

$$a_T = \frac{25}{14} \text{ AU}.$$

The spacecraft leaves Earth at the perihelion of this transfer orbit. Since the aphelion of the transfer ellipse lies beyond the orbit of Mars, the spacecraft intersects Mars' circular orbit on the outbound part of its trajectory before reaching aphelion.

Assume that Earth and Mars move in circular, coplanar orbits around the Sun. The spacecraft is launched by an instantaneous tangential burn from its circular Earth parking orbit. After leaving Earth's sphere of influence, its hyperbolic excess velocity relative to Earth is parallel to and in the same direction as Earth's heliocentric velocity.

At Mars, the spacecraft approaches along a hyperbolic trajectory. It passes through a periapsis altitude of 600 km above the Martian surface and performs an instantaneous retrograde burn to enter an elliptical orbit around Mars having an apoapsis altitude of 8600 km.

Assume that Mars is located at the outbound intersection point of its orbit with the transfer ellipse when the spacecraft arrives there.

Neglect atmospheric resistance and the gravitational effects of all bodies other than the Sun, Earth and Mars.

Use the following data. The gravitational parameter of the Sun is  $\mu_s = 1.35 \times 10^{20} \text{ m}^3 \text{ s}^{-2}$ ,

The gravitational parameters of Earth and Mars are  $\mu_E = 4.00 \times 10^5 \text{ km}^3 \text{ s}^{-2}$ ,  $\mu_M = 4.00 \times 10^4 \text{ km}^3 \text{ s}^{-2}$ ,

The mean radii of Earth and Mars are  $R_E = 6000 \text{ km}$ ,  $R_M = 3400 \text{ km}$ ,

Earth and Mars are assumed to move in circular heliocentric orbits of radii  $r_E = 1.00 \text{ AU}$ ,  $r_M = 1.44 \text{ AU}$ , where  $1 \text{ AU} = 1.50 \times 10^{11} \text{ m}$ .

Here,  $\mu = GM$  denotes the gravitational parameter of the attracting body,  $R_E$  and  $R_M$  denote the planetary radii, and  $r_E$  and  $r_M$  denote the orbital radii of Earth and Mars measured from the Sun.

$$v^2 = \mu \left( \frac{2}{r} - \frac{1}{a} \right)$$

For a conic orbit around a body of gravitational parameter  $\mu$ ,

where  $v$  is the orbital speed at distance  $r$  from the attracting body and  $a$  is the semimajor axis of the

orbit.

For an elliptical orbit of eccentricity  $e$ , semilatus rectum  $p$ , true anomaly  $\theta$ , and specific angular

$$\text{momentum } h, \quad r = \frac{p}{1 + e \cos \theta}, \quad v_r = \frac{\mu}{h} e \sin \theta, \quad v_\theta = \frac{\mu}{h} (1 + e \cos \theta),$$

where  $v_r$  and  $v_\theta$  are respectively the radial and transverse components of velocity. Also,  $p = a(1 - e^2)$ ,  $h = \sqrt{\mu p}$ .

$$v_p = \sqrt{v_\infty^2 + \frac{2\mu_p}{r_p}}$$

For a hyperbolic planetary encounter,

where  $v_\infty$  is the spacecraft's hyperbolic excess speed relative to the planet,  $\mu_p$  is the planet's gravitational parameter,  $r_p$  is the periapsis distance measured from the planet's centre, and  $v_p$  is the spacecraft speed at periapsis.

1) The tangential velocity impulse  $\Delta v_1$  required in the 250 km circular Earth parking orbit to inject the spacecraft onto the specified heliocentric transfer ellipse is closest to :

- (A) 4.00 km s<sup>-1</sup>
- (B) 4.81 km s<sup>-1</sup>
- (C) 6.00 km s<sup>-1</sup>
- (D) 12.81 km s<sup>-1</sup>

2) Calculate the retrograde impulse  $\Delta v_2$  required at the 600 km Martian periapsis to insert the spacecraft into the specified elliptical orbit. The value of  $\Delta v_2$  is closest to :-

- (A) 6.00 km s<sup>-1</sup>
- (B) 7.00 km s<sup>-1</sup>
- (C) 8.00 km s<sup>-1</sup>
- (D) 9.00 km s<sup>-1</sup>

### Common Content for Question No. 3 to 4

Three mutually perpendicular, infinite grounded conducting planes occupy  $x = 0$ ,  $y = 0$ ,  $z = 0$ . The region  $x > 0$ ,  $y > 0$ ,  $z > 0$  is vacuum. A point charge  $q = 6\mu\text{C}$  is placed at P ( $a, a, a$ ), where  $a = 30$

cm. Take  $k = \frac{1}{4\pi\epsilon_0}$ .

Using the method of images, the conducting planes may be replaced by seven image charges.

Images obtained by reflection in an odd number of planes carry charge  $-q$ , while those obtained by reflection in an even number of planes carry charge  $+q$ . The force on the real charge equals the Coulomb force due to all the image charges, whereas the electrostatic potential energy of the system

is  $U = \frac{1}{2}qV_{\text{images}}$ , where  $V_{\text{images}}$  is the potential at the position of the real charge due to all the image charges.

3) The magnitude of the net electrostatic force acting on the real charge is

- (A)  $\frac{kq^2}{a^2} \left( \frac{1}{4} - \frac{1}{4\sqrt{2}} + \frac{1}{12\sqrt{3}} \right)$
- (B)  $\frac{\sqrt{3}kq^2}{a^2} \left( \frac{1}{4} - \frac{1}{4\sqrt{2}} + \frac{1}{12\sqrt{3}} \right)$



$$(C) \frac{3kq^2}{a^2} \left( \frac{1}{4} - \frac{1}{4\sqrt{2}} + \frac{1}{12\sqrt{3}} \right)$$

$$(D) \frac{kq^2}{a^2} \left( \frac{1}{4} + \frac{1}{4\sqrt{2}} + \frac{1}{12\sqrt{3}} \right)$$

4) The charge is moved quasistatically along the line  $x = y = z$  from  $(a, a, a)$  to  $(2a, 2a, 2a)$ . The work done by the external agent during this process is

$$(A) \frac{kq^2}{4a} \left( 3 - \frac{3}{\sqrt{2}} + \frac{1}{\sqrt{3}} \right)$$

$$(B) \frac{kq^2}{8a} \left( 3 - \frac{3}{\sqrt{2}} + \frac{1}{\sqrt{3}} \right)$$

$$(C) -\frac{kq^2}{8a} \left( 3 - \frac{3}{\sqrt{2}} + \frac{1}{\sqrt{3}} \right)$$

$$(D) \frac{kq^2}{16a} \left( 3 - \frac{3}{\sqrt{2}} + \frac{1}{\sqrt{3}} \right)$$

## SECTION-II (i)

### Common Content for Question No. 1 to 2

Three concentric conductors are arranged as follows: A is a solid conducting sphere of radius  $R$ ; B is a conducting spherical shell of inner radius  $2R$  and outer radius  $4R$ ; and C is a conducting spherical shell of inner radius  $8R$  and outer radius  $16R$ . All intervening regions are vacuum. Initially, the total charges on A, B, and C are respectively  $Q$ ,  $-Q$ ,  $3Q$ . The following operations are performed successively. Electrostatic equilibrium is established after each operation.

**Stage I :** Conductor B is connected to earth while A and C remain isolated. The earth connection of B is then removed.

**Stage II :** Conductor C is connected to earth while A and B remain isolated. The earth connection of C is then removed.

**Stage III :** Conductor A is connected to earth while B and C remain isolated.

The earth is an infinitely large charge reservoir at zero potential. The connecting wires have negligible resistance and capacitance. Any decrease in electrostatic field energy during grounding is

dissipated as heat. It is given that  $\frac{1}{4\pi\epsilon_0} \frac{Q^2}{R} = 88 \text{ J}$ .

1) After Stage III reaches electrostatic equilibrium, let the charge on conductor A be  $q_A$ . Find  $100 \left( \frac{q_A}{Q} \right)$ .

2) Find the energy dissipated during stage III, in joules.

### Common Content for Question No. 3 to 4

One of the important problems in modern astrophysics is the existence and distribution of dark matter. Its presence is inferred from its gravitational effects on visible matter. Stars in a galaxy move approximately in circular orbits about the Galactic center. If most of the Galactic mass were concentrated near the center, their orbital speed would decrease with increasing distance. However,

observations show that over a considerable range the orbital speed remains nearly constant, indicating the presence of additional non-luminous matter.

Consider a simplified model of our Galaxy in which the visible mass is concentrated within a spherically symmetric core of radius  $r_1 = 4.0$  kpc. Let  $M_1$  be the total mass enclosed within  $r = r_1$ . Outside this core, stars move in concentric circular orbits with constant speed  $v_0 = 240$  km  $s^{-1}$  throughout  $r_1 \leq r \leq r_2$ , where  $r_2 = 7r_1$ .

Assume that the mass of the orbiting stars is negligible, dark matter is distributed spherically symmetrically about the Galactic center, Newtonian gravitation is applicable, and the visible-matter density in  $r_1 \leq r \leq r_2$  is negligible. Thus, all additional mass beyond  $r_1$  in this region is dark matter. Use  $G = 6.67 \times 10^{-11}$  N  $m^2$   $kg^{-2}$ , 1 kpc =  $3.086 \times 10^{19}$  m.

3) The dark-matter density in  $r_1 \leq r \leq r_2$  varies as  $\rho_2(r) \propto r^{-n}$ . Calculate  $n$ .

4) Let  $M_2$  be the total dark-matter mass contained between  $r = r_1$  and  $r = r_2$ . Calculate  $\frac{M_2}{M_1}$ .

### Common Content for Question No. 5 to 6

In a hypothetical electrostatic theory, Gauss's law for a spherically symmetric charge distribution is

modified as  $\oint_S \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} \int_V \rho(r') \left[ 3 - 10 \left( \frac{r'}{R} \right)^2 + 7 \left( \frac{r'}{R} \right)^4 \right] dV$ , where  $r'$  is the radial coordinate of the volume element  $dV$ ,  $V$  is the volume enclosed by the spherical Gaussian surface  $S$ , and  $R$  is a fixed characteristic length associated with the charged body.

The usual principles of spherical symmetry and electrostatic superposition remain valid, and the electric potential is related to the electric field by  $\vec{E} = -\nabla V$ .

A non-conducting solid sphere of radius  $R$  contains a uniform actual charge density  $\rho_0 > 0$ . There are

no other charges in space. It is given that  $\frac{\rho_0 R^2}{\epsilon_0} = 120V$ .

Take the electric potential at infinity to be zero.

5) The magnitude of the electric field inside the sphere is maximum at  $r = r_m$ . Find  $100 \left( \frac{r_m}{R} \right)$ .

6) Find the potential difference  $V(0) - V(R)$  in volts.

### SECTION-II (ii)

1) Two fixed positive point charges  $4Q$  and  $Q$  are placed on a horizontal insulating line at  $x = 0$  and  $x = 3a$ , respectively. A small bead of mass  $m$  carrying positive charge  $q$  can move without friction along the line joining the fixed charges. The bead is initially placed at the electrostatic equilibrium position between the charges. It is then displaced slightly along the line and released from rest. Assume that the displacement is sufficiently small for the resulting motion to be simple harmonic. Neglect gravitational effects and radiation losses. Given  $Q = 3.0$   $\mu C$ ,  $q = 2.0$   $\mu C$ ,  $a = 0.30$  m,  $m = 15$

$\frac{1}{g}$  and  $\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9$  N  $m^2$   $C^{-2}$ , find the angular frequency of small oscillations of the bead, in rad  $s^{-1}$ .

2) An infinitely extended non-conducting slab occupies the region  $-a \leq x \leq a$ , with the  $yz$ -plane as its central plane. Its volume charge density is  $\rho(x) = \rho_0 \left(1 - \frac{|x|}{a}\right)$ ,  $|x| \leq a$ , and is zero for  $|x| > a$ . The surrounding medium is vacuum. The half-thickness of the slab is  $a = 0.30$  m, and  $\frac{\rho_0}{\epsilon_0} = 800 \text{Vm}^{-2}$ . The electric potential is chosen to be  $V(0) = 0$ . A point P, having coordinate  $x_p > a$ , lies outside the slab on the positive  $x$ -axis. If  $V(P) = -60\text{V}$ , find the distance of P from the central plane in centimetres.

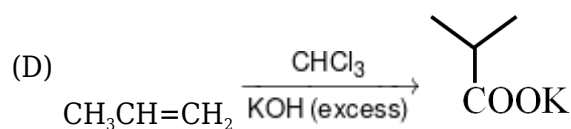
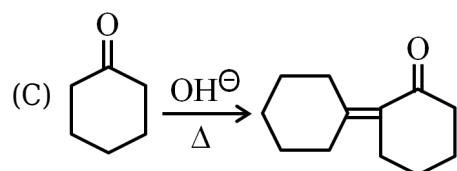
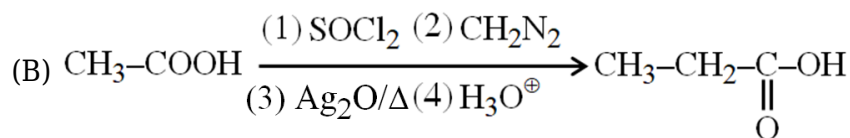
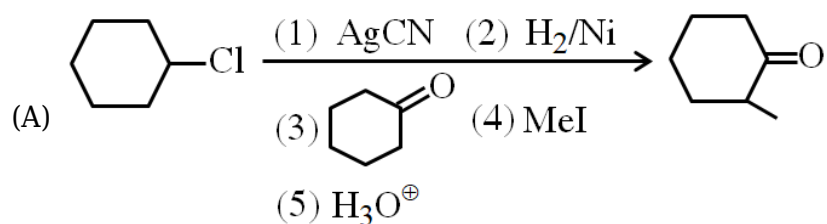
3) A non-conducting charged body occupies a finite cylindrical annulus whose axis coincides with the  $z$ -axis :  $R \leq r \leq 2R$ ,  $-L \leq z \leq L$ , where  $r$  is the perpendicular distance from the  $z$ -axis. Its volume

charge density varies as  $\rho(r, z) = \rho_0 \left(1 + \frac{r^2}{R^2} + \frac{2z^2}{L^2}\right)$ . The region  $0 \leq r < R$ ,  $-L \leq z \leq L$  is an empty coaxial cylindrical cavity containing no charge. Consider the closed Gaussian surface forming the complete boundary of the charged material. It consists of the outer curved surface  $r = 2R$ , the inner curved surface  $r = R$ , and the two annular end faces at  $z = \pm L$ . On the inner curved surface, the outward normal is directed from the charged material into the cavity. It is given that  $\pi \rho_0 R^2 L = 4q_0$ ,  $\frac{q_0}{\epsilon_0} = 1 \text{ Nm}^2\text{C}^{-1}$ . Using Gauss's law, find the total outward electric flux through the complete Gaussian surface, in  $\text{N m}^2\text{C}^{-1}$ .

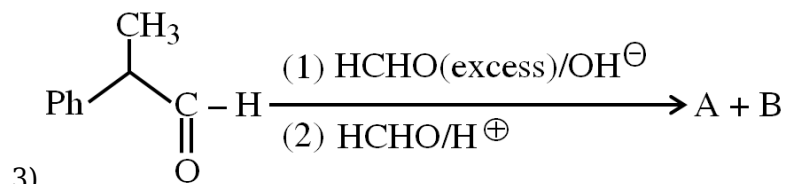
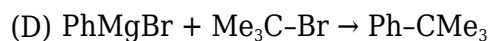
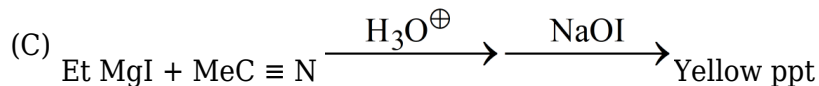
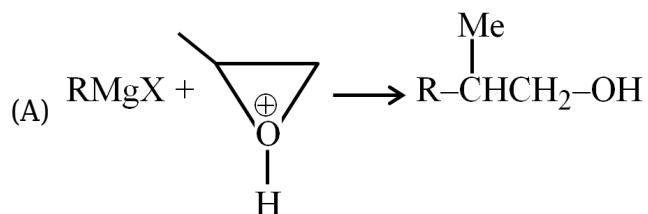
## PART-3 : CHEMISTRY

### SECTION-I (i)

1) Which of following reaction shows correct major product?



2) Select reaction(s) with correct major product-

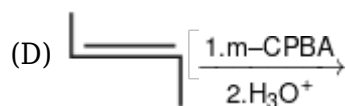
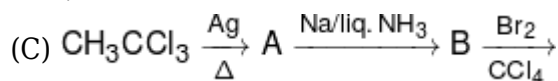
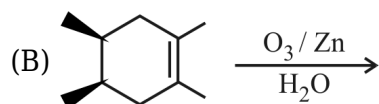
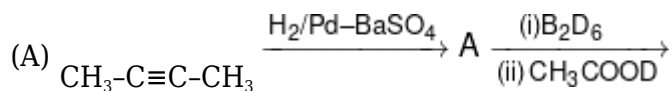


Select the incorrect statements for formation of product (A) [considering that only (B) gives +ve Tollen's test].

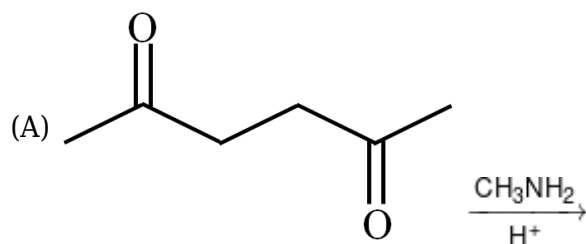
- (A) Number of stereoisomer of product (A) are (2)
- (B) Reaction follows only Aldol condensation but not Cannizaro reaction.
- (C) Reaction involves preparation of 1,3-diol in one of the steps
- (D) One of the steps of reaction follows disproportionation reaction

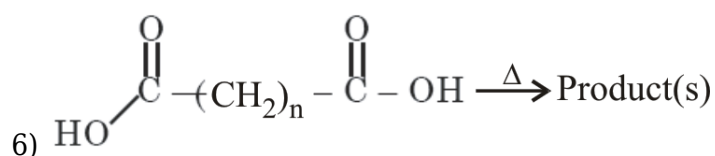
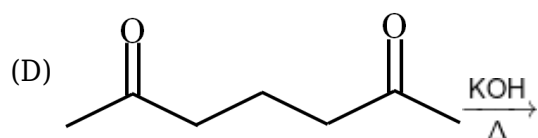
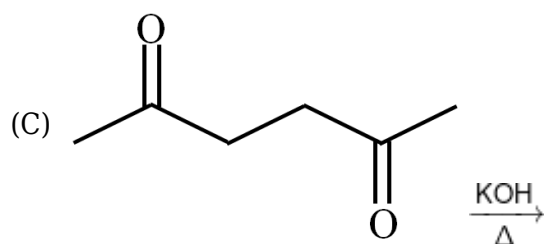
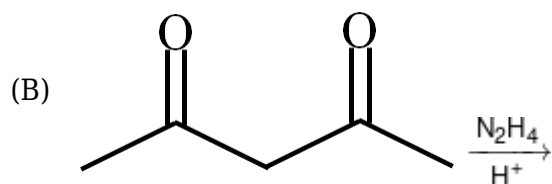
4)

In which of the following reaction final major product is meso :



5) In which of the following reaction(s) major product is/are aromatic





when  $n = 1$ , organic product formed is  $\text{P}_1$

when  $n = 3$ , organic product formed is  $\text{P}_2$

when  $n = 4$ , organic product formed is  $\text{P}_3$

Correct statement(s) is/are is :

(A)  $\text{P}_1$  gives Tollen's test

(B)  $\text{P}_2$  is cyclic anhydride

(C)  $\text{P}_3$  gives 2,4-DNP test

(D)  $\text{P}_3$  gives Tollen's test

#### SECTION-I (ii)

#### Common Content for Question No. 1 to 2

An acyclic alkene (P) has molecular formula ( $\text{C}_9\text{H}_{18}$ ). Reductive ozonolysis of (P) produces exactly two compounds (A) & (B)

(1) (A) gives a positive iodoform test.

(2) (B) does not give positive iodoform test

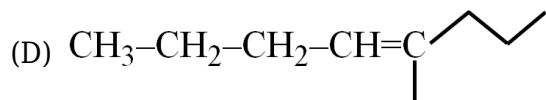
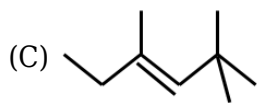
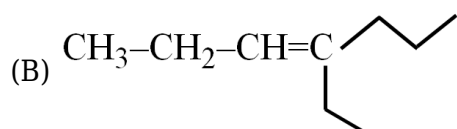
(3) (B) undergoes cannizaro reaction

(4) The alcohol obtained from the cannizaro reaction of B, on oxidation gives a carboxylic acid containing one carbon more than (A).

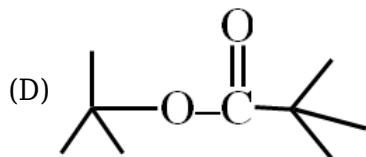
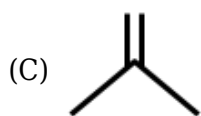
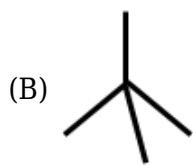
(5) (P) shows geometrical isomerism.

1) Correct structure of (P) is-

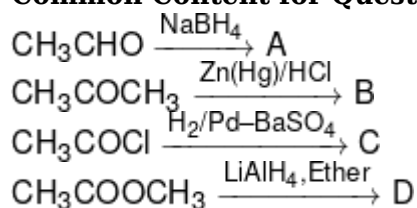




2) Carboxylic acid formed on cannizaro reaction of (B) undergoes Kolbe's electrolysis. Major product of reaction is-



**Common Content for Question No. 3 to 4**



Where A, B, C and D are the respective major product(s).

3) Which of the following correctly represent A, B, C and D, respectively?

- (A) A =  $\text{CH}_3\text{CH}_2\text{OH}$ , B =  $\text{CH}_3\text{CH}_2\text{CH}_3$ , C =  $\text{CH}_3\text{CHO}$ , D =  $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{OH}$   
 (B) A =  $\text{CH}_3\text{CH}_2\text{OH}$ , B =  $\text{CH}_3\text{CH}_2\text{OH}$ , C =  $\text{CH}_3\text{CHO}$ , D =  $\text{CH}_3\text{CH}_2\text{OH}$   
 (C) A =  $\text{CH}_3\text{COOH}$ , B =  $\text{CH}_3\text{CH}_2\text{CH}_3$ , C =  $\text{CH}_3\text{CH}_2\text{OH}$ , D =  $\text{CH}_3\text{COOH}$   
 (D) A =  $\text{CH}_3\text{CH}_2\text{OH}$ , B =  $\text{CH}_3\text{COCH}_3$ , C =  $\text{CH}_3\text{CHO}$ , D =  $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{OH}$

4) With respect to the A, B, C and D which statement is correct?

- (A) A and C give the same product on oxidation under suitable conditions

- (B) B and C belong to the same functional group  
 (C) The formation of D does not involve cleavage of C-O bond.  
 (D) A and B both contain a carbonyl group

## SECTION-II (i)

### Common Content for Question No. 1 to 2

An optically active and optically pure hydrocarbon A with MF  $C_5H_{10}$  give +ve bromine water test.

- 1) How many total monochloro derivative are possible when compound A react with  $Cl_2/h\nu$  to give free radical substitution reaction?
- 2) How many total isomers of compound A gives racemic mixture with cold & dil Alk.  $KMnO_4$  (Baeyer's reagent)

### Common Content for Question No. 3 to 4

An unknown organic compound A contains 69.77% carbon, 11.63% hydrogen and rest is oxygen. The molecular mass of A was found to be 86.

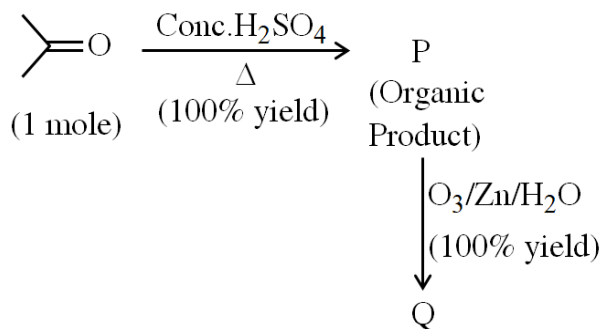
- 3) How many isomeric A do not reduce Fehling solution but form a bisulphite adduct and also give positive iodoform test?
- 4) How many isomeric A can give Cannizaro reaction but do not undergo cross aldol with  $HCHO$ ?

### Common Content for Question No. 5 to 6

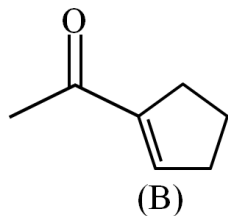


- 5) B on reaction with  $HS-CH_2-CH_2-SH$ ,  $BF_3$  followed by Raney Ni ( $Ni-H_2$ ), gives X as a major organic product. What is molar mass of X.  
 (Atomic mass : C = 12, H = 1, O = 16)
- 6) What is the total number of stereoisomer of product C.

## SECTION-II (ii)



- 1)  
What is the weight of Q (organic product) obtained in grams?
- 2) How many total number of lactides are obtained when a mixture of butanoic acid & propanoic acid is allowed to react with Red P/Br<sub>2</sub> followed by acidic hydrolysis (excess H<sub>2</sub>O) and heating?
- 3) The reactant 'A' undergoes intramolecular aldol condensation when heated in presence of NaOH to obtain the product 'B'



The number of methylene groups (-CH<sub>2</sub>-) in 'A' is



## ANSWER KEYS

### PART-1 : MATHEMATICS

#### SECTION-I (i)

| Q. | 1       | 2   | 3   | 4     | 5   | 6       |
|----|---------|-----|-----|-------|-----|---------|
| A. | A,B,C,D | B,C | B,D | A,B,C | A,C | A,B,C,D |

#### SECTION-I (ii)

| Q. | 7 | 8 | 9 | 10 |
|----|---|---|---|----|
| A. | A | C | B | A  |

#### SECTION-II (i)

| Q. | 11   | 12   | 13    | 14   | 15   | 16   |
|----|------|------|-------|------|------|------|
| A. | 1.00 | 6.00 | 10.00 | 4.00 | 3.00 | 2.00 |

#### SECTION-II (ii)

| Q. | 17 | 18 | 19 |
|----|----|----|----|
| A. | 8  | 1  | 3  |

### PART-2 : PHYSICS

#### SECTION-I (i)

| Q. | 20      | 21      | 22  | 23    | 24    | 25    |
|----|---------|---------|-----|-------|-------|-------|
| A. | A,B,C,D | A,B,C,D | B,C | A,B,C | A,B,C | A,B,C |

#### SECTION-I (ii)

| Q. | 26 | 27 | 28 | 29 |
|----|----|----|----|----|
| A. | B  | C  | B  | B  |

#### SECTION-II (i)

| Q. | 30    | 31   | 32   | 33   | 34    | 35    |
|----|-------|------|------|------|-------|-------|
| A. | 45.45 | 9.00 | 2.00 | 6.00 | 44.72 | 20.00 |

#### SECTION-II (ii)

| Q. | 36 | 37 | 38  |
|----|----|----|-----|
| A. | 20 | 60 | 100 |

### PART-3 : CHEMISTRY

## SECTION-I (i)

| Q. | 39      | 40  | 41    | 42      | 43  | 44  |
|----|---------|-----|-------|---------|-----|-----|
| A. | A,B,C,D | B,C | A,B,D | A,B,C,D | A,B | B,C |

## SECTION-I (ii)

| Q. | 45 | 46 | 47 | 48 |
|----|----|----|----|----|
| A. | C  | D  | A  | A  |

## SECTION-II (i)

| Q. | 49   | 50   | 51   | 52   | 53     | 54   |
|----|------|------|------|------|--------|------|
| A. | 4.00 | 6.00 | 2.00 | 1.00 | 138.00 | 4.00 |

## SECTION-II (ii)

| Q. | 55 | 56 | 57 |
|----|----|----|----|
| A. | 72 | 10 | 4  |

# SOLUTIONS

## PART-1 : MATHEMATICS

1) (A) POI is (2, 1) and (-2, -1)

$$m_1 = \frac{-1}{2} \quad m_2 = \frac{3}{1}$$

(B)  $m_1 m_2 = -1$

(C)  $2x^4 + x^2 - 3 = 0$  POI (1, 1) ; (-1, 1)

$$x^2 = 1$$

$$x = \pm 1$$

$$\left. \frac{dy}{dx} \right|_1 = 2x$$

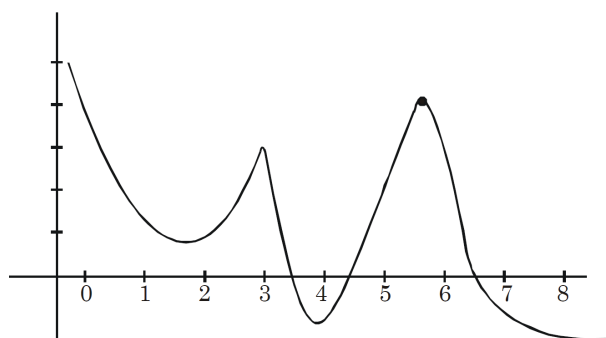
$$\left. \frac{dy}{dx} \right|_2 = \frac{-x}{2y}$$

$$2) (f(x))^3 + x^3 + (-f(x))^3 - 3f(x)x(-f(x)) = 0$$

$$\Rightarrow f(x) = -f(x) = x \text{ (Rejected)}$$

$$\text{or } f(x) - f(x) + x = 0$$

$$\Rightarrow f(x) = (x + 1) + ce^x$$



3)

$$f(x) = f(x) (f(x))^2$$

$$\Rightarrow f(x) = 0, f(x) = 1, f(x) = -1$$

$f(x) = 0$  for atleast 4 points

$f(x) = 1$  for atleast 5 points

$f(x) = -1$  for atleast 2 point

so  $f(x) = f(x) (f(x))^2$  for atleast 11 points.

$$(e^{2x} \cdot f(x) \cdot f(x))' = 0$$

$$\Rightarrow f(x) = 0 \text{ has at least 3}$$

$$\mu = 6 ; \lambda = 11$$

$$4) \alpha = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^6 + \dots + \alpha = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}$$

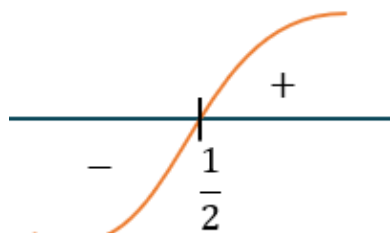
$$\therefore g(x) = 2^{\frac{x}{3}} + 2^{\left(\frac{1}{3}\right)(1-x)}$$

$$\therefore g(x) = 2^{\frac{x}{3}} + 2^{\frac{1-x}{3}}$$

where  $g(0) = 1 + 2^{\frac{1}{3}}$  &  $g(1) = 1 + 2^{\frac{1}{3}}$

$$\therefore g'(x) = \frac{1}{3} \left( 2^{x/3} - \frac{2^{1/3}}{2^{x/3}} \right) = 0$$

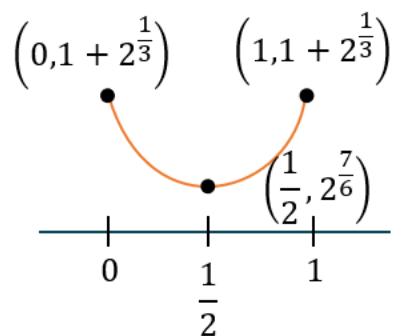
$$\Rightarrow 2^{\frac{2x}{3}} = 2^{\frac{1}{3}} \Rightarrow x = \frac{1}{2} = \text{critical point}$$



graph of  $g'(x) =$

$$\& g\left(\frac{1}{2}\right) = 2^{\frac{7}{6}}$$

$\therefore$  graph of  $g(x)$  in  $[0, 1]$



5)

$$f(x) - 2 = (x - 1)^2(x - 4)^3$$

$$f(2) - 2 = -8 \Rightarrow f(2) = -6$$

$$f(3) - 2 = -4 \Rightarrow f(3) = -2$$

$$f(2) + f(3) = -8$$

$$f'(x) = (x - 1)(x - 4)^2(5x - 11)$$

$\Rightarrow f(x)$  has local minima

$$\text{at } x = \frac{11}{5}$$

$$f(x) = \begin{cases} (x+1)^7 - 2x^2 & , \quad x < 0; \\ x^2 - x + 1 & , \quad 0 \leq x < 1; \\ \frac{2}{3}x^3 - 4x^2 + 7x - \frac{8}{3} & , \quad 1 \leq x < 3; \\ (x-2) \log_e (x-2) - x + \frac{10}{3} & , \quad x \geq 0 \end{cases}$$

6)

for  $x < 0$ ,  $f(x)$  is continuous

$$\text{and } \lim_{x \rightarrow -\infty} f(x) = -\infty \quad \text{and } \lim_{x \rightarrow 0^-} f(x) = 1$$

Hence,  $(-\infty, 1) \subset \text{Range of } f(x) \text{ in } (-\infty, 0)$

$$f'(x) = 7(x+1)^6 - 2x^2 \text{ which changes sign in } (-\infty, 0)$$

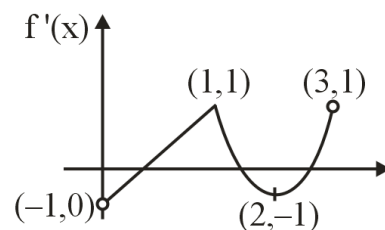
$\Rightarrow f(x)$  is non-monotonic in  $(-\infty, 0)$

For  $x \geq 3$ ,  $f(x)$  is again continuous and  $\lim_{x \rightarrow \infty} f(x) = \infty$  and  $f(3) = \frac{1}{3}$

$$\Rightarrow \left[ \frac{1}{3}, \infty \right) \subset \text{Range of } f(x) \text{ in } [3, \infty)$$

$$f'(x) = \begin{cases} 2x-1 & , 0 \leq x < 1 \\ 2x^2-8x+7 & , 1 \leq x < 3 \end{cases}$$

Hence  $f'$  has a local maximum at  $x = 1$  and  $f'$  is NOT differentiable at  $x = 1$



$$7) a_1 = 0 \text{ and } \frac{1}{1-a_{n+1}} - \frac{1}{1-a_n} = 2n+1$$

$$\Rightarrow a_n = \frac{n^2-1}{n^2}$$

$$\prod_{n=2}^{\infty} a_n = \lim_{n \rightarrow \infty} \frac{(2^2-1)}{2^2} \frac{(3^2-1)}{3^2} \frac{(4^2-1)}{4^2} \dots \frac{(n^2-1)}{n^2}$$

$$= \lim_{n \rightarrow \infty} \left[ \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \dots \frac{(n-1)}{n} \right] \left[ \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} \dots \frac{n+1}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left( \frac{n+1}{2} \right) = \frac{1}{2}$$

$$8) b_n = \sqrt{\frac{n+1}{n}} - \sqrt{\frac{n+2}{n+1}}$$

$$9) a_n = \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$$

$$\square S_n = 1 - \frac{1}{\sqrt{n+1}}$$

$S_n$  is rational if and only if  $(n+1)$  is perfect square

$$44 < \sqrt{2015} < 45$$

$$\square n+1 = 2^2, 3^2 \dots 44^2$$

$\square$  43 possible values of  $n$  are there

$$10) 1 - S_n = \frac{1}{\sqrt{n+1}}$$

$$(1 - S_n)^{-2} = \left( \sqrt{n+1} \right)^2$$

$$T_n = n+1$$

$$\sum T_n = \sum n + \sum 1$$

$$= \frac{n(n+1)}{2} + n$$

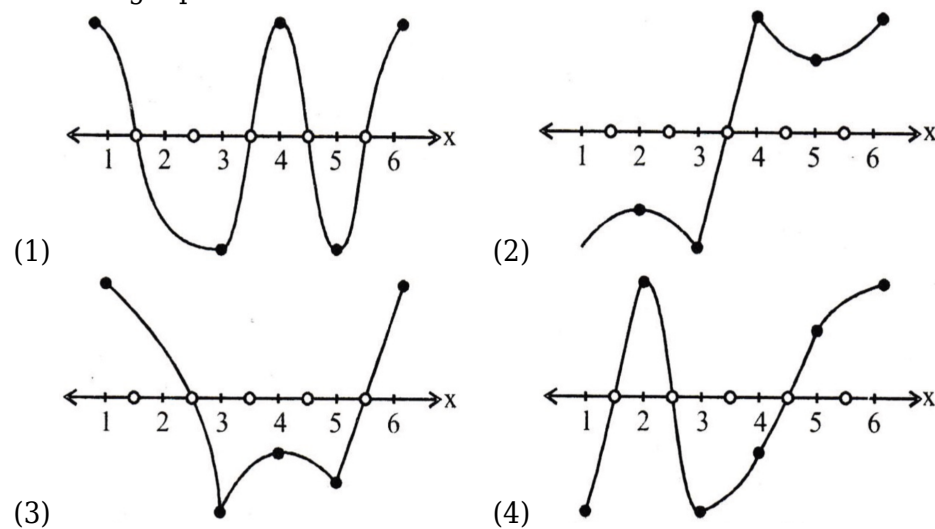
$$\sum T_n = n \left( \frac{n+3}{2} \right)$$

put  $n = 10$ ,  $\frac{10 \times 13}{2} = 65$

11)  $\square g(x) = e^{f(x)} \Rightarrow g'(x) = e^{f(x)} f'(x) \forall x \in \mathbb{R}$   
 $\Rightarrow g'(x) = e^{f(x)} [2018 \cdot (x - 2019)^3 \cdot (x - 2020)^4 \cdot (x - 2021)^5 \cdot (x - 2022)^6]$   
 $\square f'(x)$  changes its sign from negative to positive in the neighbourhood of  $x = 2021$   
 $\Rightarrow f(x)$  has local minima at  $x = 2021$   
 so, the number of local minima is one.

12)  $h(x) = f(x)f'(x)$   
 $h'(x) = (f(x)f'(x))'$   
 $f'(x) = 0$  has 7 distinct real roots which contain  $f(x) = 0$   
 $\Rightarrow h'(x) = 0$  has 6 distinct roots

13)  $f(6) > 0$ ,  $f(3) < 0$ ,  $f(2)$ ,  $f(4)$  will be of opposite sign.  $f(1)$ ,  $f(5)$  will be opposite sign.  
 Following 4 possibilities are as

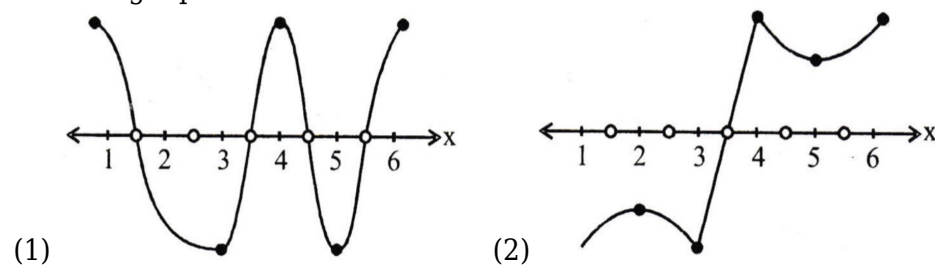


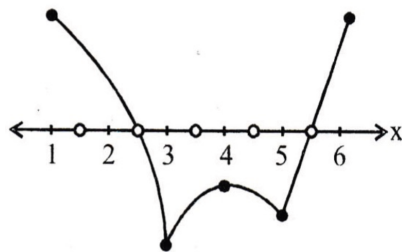
Number of possible roots 1, 2, 3, 4.

$A = \{1, 2, 3, 4\}$

Hence,  $\sum n(A) = 10$

14)  $f(6) > 0$ ,  $f(3) < 0$ ,  $f(2)$ ,  $f(4)$  will be of opposite sign.  $f(1)$ ,  $f(5)$  will be opposite sign.  
 Following 4 possibilities are as



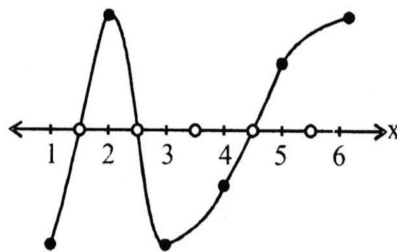


(3)

Number of possible roots 1, 2, 3, 4.

$$A = \{1, 2, 3, 4\}$$

$$\text{Hence, } \sum n(A) = 10$$



(4)

$$15) f'(x) < 0$$

$$\square f(x^3 + f(x)) \leq f(-f(x) - x^3)$$

$$\Rightarrow x^3 + f(x) \geq -f(x) - x^3$$

$$\Rightarrow f(x) + x^3 \geq 0$$

$$\Rightarrow 3x^2 + 6x - 1 \leq 0$$

$$\text{As } x \in \mathbb{Z}, x \in \{-2, -1, 0\}$$

$$16) f'(x) < 0$$

$$\square f(x^3 + f(x)) \leq f(-f(x) - x^3)$$

$$\Rightarrow x^3 + f(x) \geq -f(x) - x^3$$

$$\Rightarrow f(x) + x^3 \geq 0$$

$$\Rightarrow 3x^2 + 6x - 1 \leq 0$$

$$\text{As } x \in \mathbb{Z}, x \in \{-2, -1, 0\}$$

$$17) N = \frac{9 \cdot (10^{16} - 1) - 5(10^8 - 1)}{9 \cdot 10^8 + 4}$$

$$= \frac{9 \cdot 10^{16} - 5 \cdot 10^8 - 4}{9 \cdot 10^8 + 4}$$

$$= \frac{(9 \cdot 10^8 + 4)(10^8 - 1)}{9 \cdot 10^8 + 4}$$

$$(\square 9x^2 - 5x - 4 = (9x + 4)(x - 1))$$

$$= 10^8 - 1 = \underbrace{111 \dots 1}_{8 \text{ digits}}$$

$$18) f(x) = (b - 1)(b - 2)(-2)\sin(2x) + (b - 1)$$

$$= (b - 1)[1 - 2(b - 2)\sin 2x] \neq 0$$

$$b \in \left(\frac{3}{2}, \frac{5}{2}\right)$$

$$b \neq 1,$$

19)

$$\text{Put } x = 2\cos\theta, y = 2\sin\theta$$

$$E = \frac{\sin\theta + 1}{\cos\theta + \sin\theta - 2}$$

$$\begin{aligned} \text{Put } \tan \frac{\theta}{2} &= t \\ E &= \frac{\frac{2t}{1+t^2} + 1}{\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} - 2} \\ \Rightarrow t^2(3E+1) + t(-2E+2) + E+1 &= 0 \\ D &\geq 0 \\ \Rightarrow E &\in [-3, 0] \Rightarrow \alpha = -3 \end{aligned}$$

## PART-2 : PHYSICS

20)

Correct Answer is (A,B,C,D)

21)

Correct Answer is (A,B,C,D)

22)

Correct Answer is (B,C)

23)

Correct Answer is (A,B,C)

24)

Correct Answer is (A,B,C)

25)

Correct Answer is (A,B,C)

26)

Correct Answer is (B)

27)

Correct Answer is (C)

28)

Correct Answer is (B)



29)

Correct Answer is (B)

30)

Correct Answer is (45.45)

31)

Correct Answer is (9.00)

32)

Correct Answer is (2.00)

33)

Correct Answer is (6.00)

34)

Correct Answer is (44.72)

35)

Correct Answer is (20.00)

36)

Correct Answer is (20)

37) Correct Answer is (60)

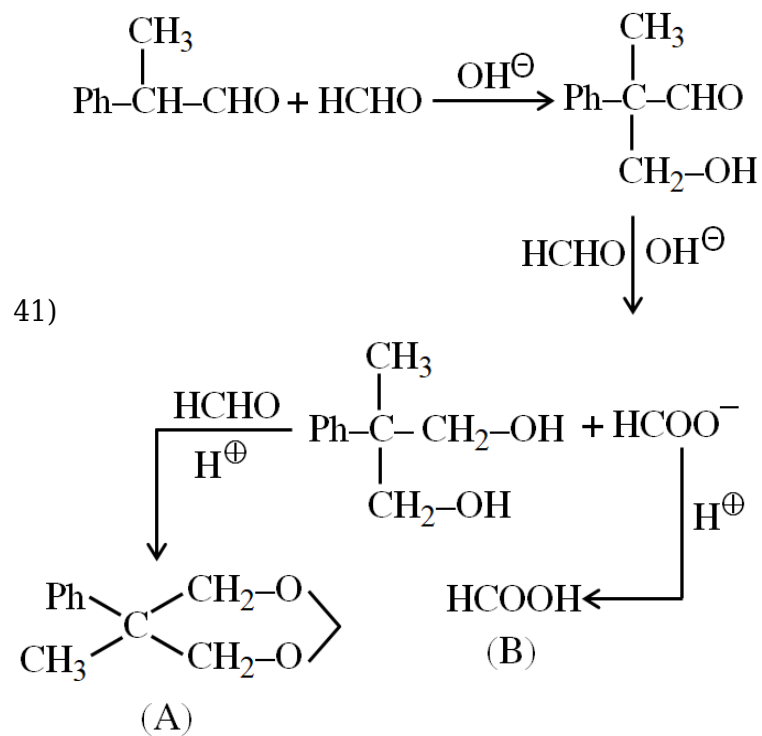
38)

Correct Answer is (100)

### PART-3 : CHEMISTRY

39) The correct answer is option (A), (B), (C), (D)

40) The correct answer is option (B), (C)



42)

The correct answer is option (A), (B), (C), (D)

43) (A) and (B) are Paal - Knorr reaction  
(C) and (D) are aldol condensation

44)

The correct answer is option (B, C)

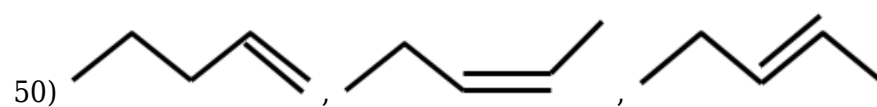
45) The correct answer is option (C)

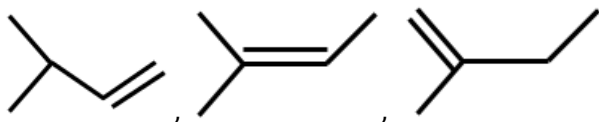
46) The correct answer is option (D)

47) The correct answer is option (A)

48) The correct answer is option (A)

49) The correct answer is 4.00

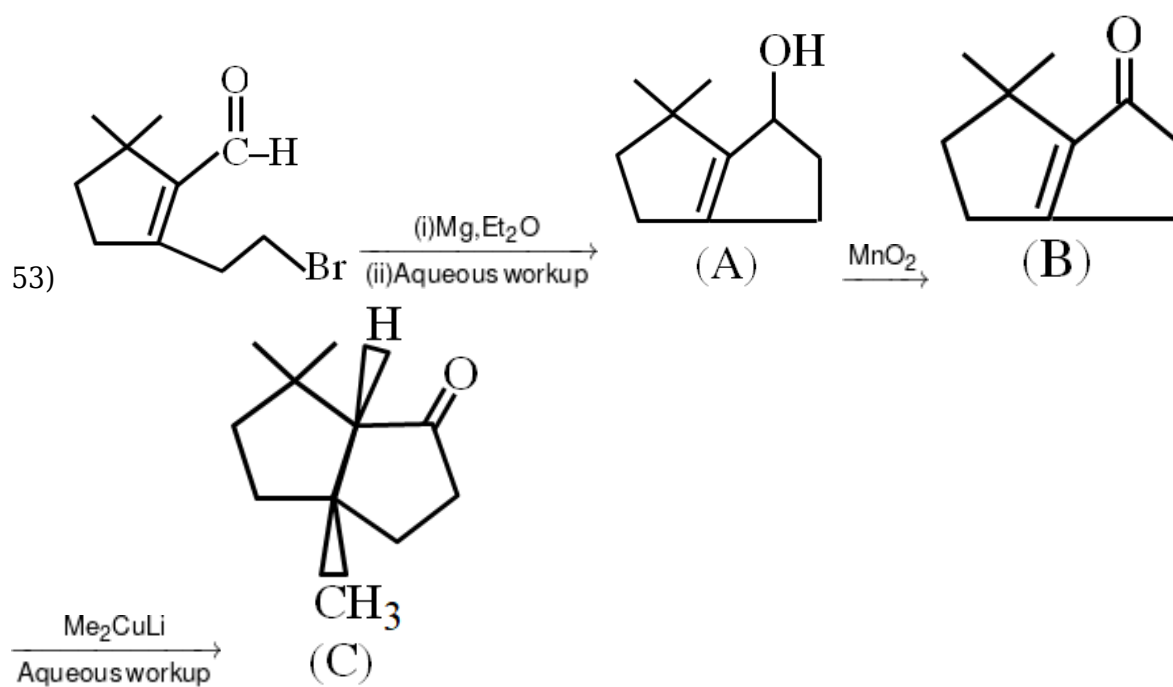
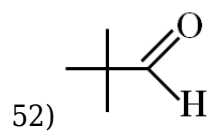
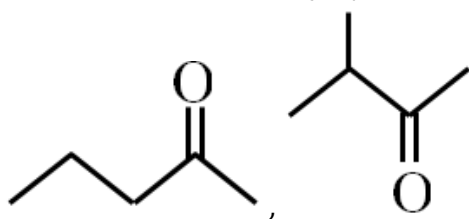


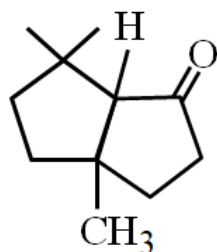
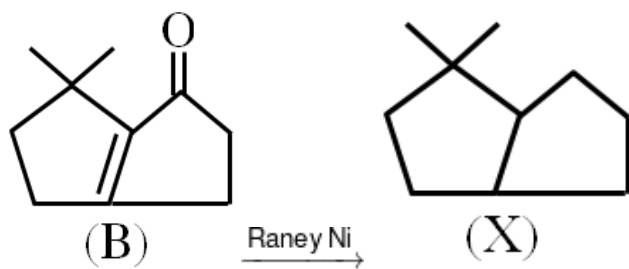


51) Empirical formula calculation

| Element | %      | Relative number | Simplest Ratio |
|---------|--------|-----------------|----------------|
| C       | 69.77% | 5.81            | 5              |
| H       | 11.63% | 11.63           | 10             |
| O       | 18.60% | 1.16            | 1              |

Molecular formula is  $C_5H_{10}O$



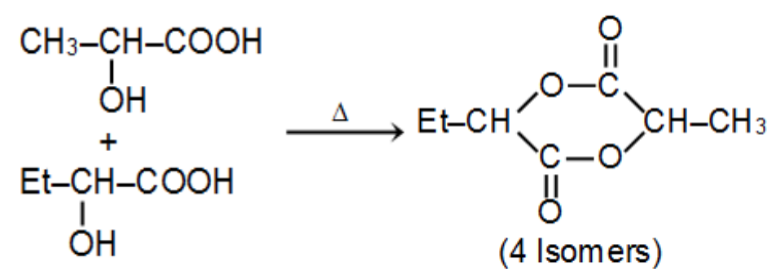
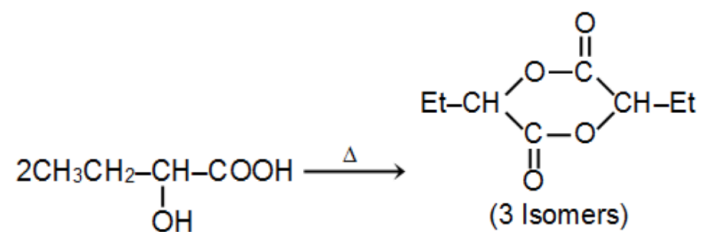
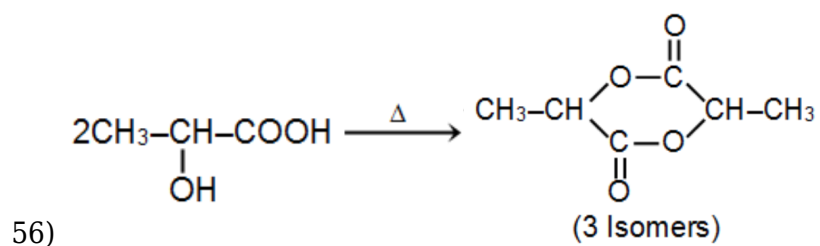


54)

cis = 2

trans = 2

55) The correct answer is 72.



57)

